

• Section 4.4 Orthogonal Bases & Gram-Schmidt

~~When  $A^T A$  is orthogonal~~

When basis vectors are orthonormal,  $A^T A$  collapses to  $I$ , eliminating matrix inversions from projection and least squares.

• The Gram-Schmidt process constructs orthonormal bases from any independent set, and the factorization  $A = QR$  is the result.

• Orthonormal vectors and  $Q$ :

Vectors  $q_1, \dots, q_n$  are orthonormal when:

$\hookrightarrow \boxed{q_i^T q_j = 0} \quad (i \neq j) \text{ and } \boxed{q_i^T q_i = 1}$

- A matrix  $Q$  with orthonormal columns satisfies  $Q^T Q = I \rightarrow$  regardless of shape. only diagonals are relevant.
- When  $Q$  is also square,  $Q^T = Q^{-1}$  and  $Q$  is called an orthogonal matrix.
- Orthogonal matrices preserve lengths:  $\|Qx\| = \|x\|$
- Examples: rotation, permutation, and reflection matrices ( $Q = I = 2uu^T$ ) are orthogonal.

## • Projection with orthonormal columns

Replacing  $A$  with  $Q$  in the projection formula:  $P = A(A^T A)^{-1} A^T$

$$\hat{x} = Q^T b \quad (\text{no matrix to invert, just dot products})$$

$$p = Q Q^T b$$

$$p = Q Q^T$$

↳ When  $Q$  is square,  $Q Q^T = I$  and  $p = b$  (projection onto the whole space).

## • Gram-Schmidt

Goal: produce orthonormal  $q_1, q_2, q_3$  from independent  $a, b, c$ .

Method: at each step, subtract from the new vector all its projections onto the directions already found, then normalise.

$$A = a$$

$$B = b - \underbrace{(q_1^T b) q_1}_{\text{projection of } b \text{ onto } a} \Rightarrow p \quad \left( \begin{array}{l} \text{subtract projection onto } A \\ \text{subtract projections onto } A \text{ \& } B \end{array} \right)$$

$$C = c - \underbrace{(q_1^T c) q_1}_{\text{projection of } c \text{ onto } a} - \underbrace{(q_2^T c) q_2}_{\text{projection of } c \text{ onto } b}$$

$$q_1 = \frac{A}{\|A\|} \quad ; \quad q_2 = \frac{B}{\|B\|} \quad ; \quad q_3 = \frac{C}{\|C\|}$$

- Each new vector uses the already normalised directions.
- The error vector (from Section 4.2) perpendicular to previous directions, is isolated at each step.

$$A = QR$$

Gram-Schmidt shows  $a, b, c$  are combinations of  $q_1, q_2, q_3$  with upper triangular coefficients.

• Resulting factorization:

$$A = QR \quad \text{where } R = Q^T A \quad (\text{upper triangular, positive diagonal})$$

• Least Squares with  $A = QR$  simplifies to back-substitution:

$$A^T A \hat{x} = A^T b \rightarrow R \hat{x} = Q^T b$$

• Section 5.1 Properties of Determinants

$$\rightarrow A^T A = R^T R$$

lower  $\Delta$     upper  $\Delta$

The determinant is a single number encoding whether  $A$  is invertible, what the pivots are, and how volumes scale under the linear map  $A$ .

• The three defining rules

Rule 1:  $\det I = 1$

Rule 2: Sign reverses on any row exchange

Rule 3:  $\det$  is linear in each row separately (other rows fixed)

• Linearity in one row means: multiplying one row by  $t$  multiplies  $\det A$  by  $t$ . Adding two versions of one row adds the determinants.

Crucially,  $\{\det(2A) = 2^n \det A, \text{ not } 2 \det A\}$

## • Rules 4 - 10 (derived)

Rule 4: Equal rows  $\rightarrow \det A = 0$

Rule 5: Elimination (row subtraction) leaves  $\det A$  unchanged.

Rule 6: Zero row  $\rightarrow \det A = 0$

Rule 7: Triangular  $A$ :  $\det A = \text{product of diagonal entries}$ .

Rule 8:  $A$  singular  $\leftrightarrow \det A = 0$ ;  $A$  invertible  $\leftrightarrow \det A \neq 0$

Rule 9:  $\{ \det(AB) = \det(A)\det(B) \}$

Rule 10:  $\{ \det(A^T) = \det A \}$

## • Computing $\det A$ :

The practical method is elimination to upper triangular  $U$ :  
 $\det A = \pm (\text{product of pivots})$

The sign is  $+1$  if the number of row exchanges is even,  $-1$  if odd.

With  $PA = LU$ :  $\det A = \pm \det U$  since  $\det L = 1$  and  $\det P = \pm 1$

Consequences of Rule 9:

$$\{ \hookrightarrow \det(A^{-1}) = \frac{1}{\det(A)} \}$$

$$\{ \hookrightarrow \det(A^T) = \det(A) \} \rightarrow \text{all row rules apply equally to columns.}$$

## • Section 5.2 Permutations and Cofactors

Two explicit formulas for  $\det A$  that avoid elimination:  
the Big Formula (sum over all  $n!$  permutations) and the  
the Cofactor Formula (recursive reduction to size  $n-1$ ).

### • The Big Formula

- Every term in  $\det A$  picks one entry from each row and one entry from each column.
- There are  $n!$  such choices.
- Each term carries the sign of its permutations:

$$\det A = \sum (\det P) \cdot a_{1\alpha} \cdot a_{2\beta} \cdots a_{n\omega}$$

- ↳ where  $(\alpha, \beta, \dots, \omega)$  ranges over all column permutations.
- ↳ Even permutations (even number of exchanges to reach the identity) get  $+1$
- ↳ Odd permutations get  $-1$
- ↳ Half of all  $n!$  permutations are even, half odd.

e.g. For  $n=3$ : six terms (three  $+$ , three  $-$ )

### • Cofactors

The cofactor  $C_{ij}$  is obtained by deleting row  $i$  and column  $j$ , then applying a sign.

$$C_{ij} = (-1)^{(i+j)} \cdot \det M_{ij}$$

- ↳ where  $M_{ij}$  is the  $(n-1) \times (n-1)$  submatrix with row  $i$  / column  $j$  removed
- ↳ The sign pattern alternates in a checkerboard  $\begin{bmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{bmatrix}$

• Cofactor expansion along any row  $i$ :

$$\det A = a_{i1}C_{i1} + a_{i2}C_{i2} + \dots + a_{in}C_{in}$$

• Or down any column  $j$ :

$$\det A = a_{1j}C_{1j} + a_{2j}C_{2j} + \dots + a_{nj}C_{nj}$$

→ Expand along the row or column with the most zeroes to minimize computation.

• Pivot Formula from Submatrices

The  $k$ th pivot equals the ratio of consecutive leading submatrix determinants:

$$d_k = \frac{\det A_k}{\det A_{k-1}}$$

↳ where  $A_k$  is the  $k \times k$  upper left matrix

↳ this holds when no row exchanges are required.

• Key Relationship

Big Formula & Cofactor Formula are both equivalent to three defining rules from Section 5.1.

→ Two ways to unpack the same object.

Practically: → Pivots (elimination) is fastest

→ Cofactors useful when rows or columns are sparse

## Section 4.4 con.

### • Inverting matrices with ~~are~~ orthogonal columns

When a matrix  $A$  has orthogonal columns,  $A^T A$  is diagonal (C)

$$A^T A = D$$

↳ The diagonal entries are the squared lengths of each column

$$(A^T A)_{ii} = \|a_i\|^2$$

↳ Off-diagonal entries are dot products between different columns (zero by orthogonality).

### • Three Cases (Previous heading)

• Orthonormal columns  $\rightarrow$  all lengths equal 1  
 $A^T A = I \quad A^T = A^{-1} \quad A^{-1} = A^T$

• Orthogonal columns  $\rightarrow$  equal lengths  $c$   
 $A^T A = c^2 I \quad A^{-1} = (\frac{1}{c^2}) A^T$

• Orthogonal columns  $\rightarrow$  varying lengths  
 $A^T A = D = \text{diag}(d_1, d_2, \dots, d_n)$

↳  $D^{-1}$  immediate:  $D^{-1} = \text{diag}(\frac{1}{d_1}, \frac{1}{d_2}, \dots, \frac{1}{d_n})$

↳ From  $A^T A = D$  (mult  $A^{-1} \rightarrow$  left;  $D^{-1} \rightarrow$  right)  
∴  $A^{-1} = D^{-1} A^T$

\*  $\rightarrow$  Orthogonality decouples columns  $\rightarrow A^T A = D$   
 $\rightarrow$  Inverse becomes trivial